



ENGG100 - Practical 8

Engineering Computing and Analysis (University of Wollongong)



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ENGG100 – Engineering Computing and Analysis

Practical 8 – Week 8

Kinematics – Rectilinear Motion

In this Practical class, you will work through the following engineering mechanics problems, focussed on kinematic rectilinear motion.

The questions listed below will take more than 2 hours to complete but are a good selection of rectilinear motion problems for you to work through.

The questions marked with an asterisk (*) will be used in Workshop 8, so make sure to complete these questions as a minimum.

***STEP 1 – General equations.**

1. The position of a particle travelling along a straight line is given by the equation $s = 1.2t^3 - 16.2t^2 + 50.4t$ m, where t is in seconds. Determine
 - a. the position of the particle when $t = 10$ sec and
 - b. the total distance it travels during the 10 sec time interval.Hint: draw the path to determine the total distance travelled.

STEP 2 – General equations.

2. The motion of a particle is defined by the relation $x = t^3 - 6t^2 + 9t + 5$, where x is expressed in metres and t in seconds. Determine
 - a. when the velocity is zero and
 - b. the position, acceleration and total distance travelled when $t = 5$ sec.

***STEP 3 – General equations.**

3. The position of a speedboat travelling in a straight line during the time interval $t = 2$ sec to $t = 10$ sec is given by $s = 4t + 1.6t^2 - 0.08t^3$ m. Determine
 - a. the boat's velocity and acceleration at $t = 4$ sec and
 - b. the boat's maximum velocity during this time interval and when does it occur?

STEP 4 – Constant acceleration equations.

4. A car can have an acceleration and a deceleration of 5 m/s^2 . If the car starts from rest and can have a maximum speed of 60 m/s , determine the shortest time it can travel a distance of 1200 m when it stops.
Hint: draw a v-t graph to help visualise the motion.

STEP 5 – Constant acceleration equations.

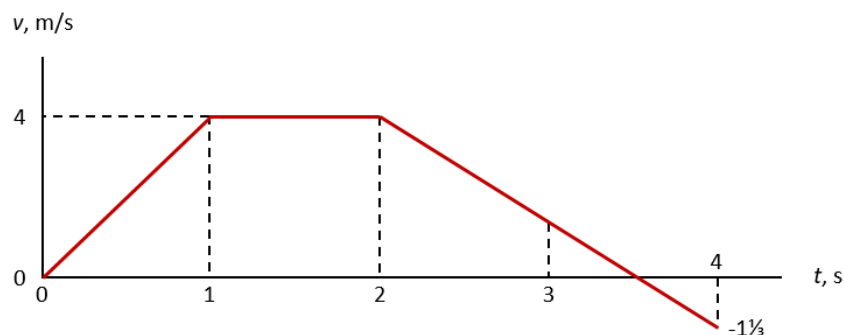
5. A cyclist starts from rest and after travelling along a straight path a distance of 20 m reaches a speed of 30 km/h. Determine
- their acceleration (if it is assumed constant) and
 - how long it takes to reach the speed of 30 km/h.

*STEP 6 – Constant acceleration equations.

6. A person standing on top of a tower throws a ball straight up (released at a height of 10 m from the ground). If the ball falls past them 0.8 seconds later, determine
- the velocity at which the ball was thrown,
 - the time of flight of the ball and
 - the velocity of the ball the instant it hits the ground.
- Hint: draw a simple diagram to visualize the motion of the ball.

*STEP 7 – Erratic motion.

7. A particle starts from rest at $x = -3$ m and moves along the x -axis with the velocity history shown.
- Plot the acceleration-time graph for the 4 seconds.
 - Plot the displacement-time graph for the 4 seconds.
 - Find the time t when the particle crosses the origin.



STEP 8 – Erratic motion.

8. A subway train travels between two of its station stops with the acceleration schedule shown. Determine
- the time interval Δt during which the train brakes to a stop with a deceleration of 2 m/s^2 , and
 - find the distance between the train stations.

